Sebastian Tamayo

Senior Software Engineer

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Accomplished Senior Web Developer with over a decade of experience designing, developing, and optimizing scalable, high-performance web applications. Specializes in building intuitive, mobile-first interfaces using Vue.js and architecting secure, extensible backend systems with Java Spring Boot and RESTful APIs.

Expert across the full software development lifecycle—from system architecture and relational/noSQL data modeling to CI/CD automation, testing strategies, and cloud-native deployments on AWS and Kubernetes. Thrives in Agile, cross-functional teams, consistently delivering user-focused, business-aligned solutions through rapid iteration and stakeholder collaboration.

Renowned for writing clean, maintainable, and testable code while mentoring engineering teams and upholding best practices rooted in SOLID principles, clean architecture, and Test-Driven Development (TDD). Proficient in modern design patterns, frontend performance optimization, and scalable microservice architectures that ensure long-term reliability, maintainability, and performance.

Driven by a passion for innovation and continuous learning, with deep technical expertise in modern JavaScript frameworks, the Java ecosystem, and DevOps tooling. Committed to delivering high-impact, real-world software solutions that elevate user experience and create tangible business value.

PROFESSIONAL EXPERIENCE

UST

Senior Software Engineer

June 2022 – June 2025 (3 years)

- Collaborated within a cross-functional Agile Scrum team, mentoring junior engineers and leading rigorous code reviews with a focus on SOLID principles, design patterns, and clean architecture, ensuring scalable, maintainable codebases across frontend and backend systems.
- Revamped a high-traffic, customer-facing application by developing a robust Vue.js frontend with responsive design, WCAG-compliant accessibility, and real-time user interactions. Integrated live auction streaming via AWS IVS and external systems like the Shopify Storefront API, increasing engagement and conversion rates.
- Built scalable backend services using Java Spring Boot, exposing RESTful APIs and integrating with Spring Data JPA and PostgreSQL for efficient data access, transactional consistency, and schema design.
- Architected a modular, microservices-based backend deployed on Amazon EKS (Kubernetes). Used Apache Kafka and Spring Cloud Stream to enable asynchronous communication between distributed Vue frontends and backend services, enhancing resilience and throughput.
- Led the development of an AI-powered chatbot using AWS Lex, AWS Lambda, and Amazon DynamoDB, providing context-aware, intelligent customer service automation that scaled with demand.
- Boosted frontend performance through lazy loading, dynamic imports, and component-level memoization in Vue, while optimizing backend performance with asynchronous Spring APIs, in-memory caching (Redis), and efficient database queries—resulting in a 30% improvement in load time and user satisfaction.
- Built and maintained CI/CD pipelines using GitHub Actions and Jenkins, integrating unit tests with JUnit, Vue Test Utils, and Cypress for E2E testing. This improved test coverage, accelerated feedback loops, and shortened deployment cycles by 22%.

Tech Stack: Vue.js, Vuex (or Pinia), TypeScript, HTML5, SCSS, Java, Spring Boot, Spring WebSocket, REST API, Spring Data JPA, PostgreSQL, DynamoDB, Redis, Apache Kafka, Spring Cloud Stream, Shopify Storefront API, AWS EKS, AWS Lambda, AWS IVS, AWS Lex, GitHub Actions, Jenkins, Vue Test Utils, Cypress, JUnit, Webpack, Vite, WCAG, Figma, Postman, SEO, TDD, CI/CD

Gorilla Logic

Full Stack Engineer

February 2018 – May 2022 (4 years 4 months)

- Designed and implemented responsive, mobile-first interfaces using Vue.js, TypeScript, and SCSS, driving a 40% increase in mobile user retention and a 30% uplift in overall platform engagement.
- Developed scalable, reusable components for enterprise dashboards, leveraging Pinia (or Vuex) for centralized state management, enabling consistency and maintainability across multiple distributed development teams.
- Built secure, high-performance backend services using Java Spring Boot, delivering well-structured RESTful APIs and integrating Spring Data JPA for seamless database interactions and domain-driven design.
- Enabled real-time dynamic content delivery by implementing WebSocket endpoints using Spring WebSocket and Server-Sent Events (SSE), enhancing interactivity for live features and eliminating full-page reloads.
- Deployed containerized microservices using Docker on AWS ECS, and automated infrastructure provisioning via AWS CloudFormation, resulting in faster release cycles and more resilient deployment environments.
- Optimized backend performance through efficient PostgreSQL queries, schema normalization, and indexing, reducing API response times by 15% under peak load conditions.
- Implemented asynchronous communication patterns using Apache Kafka (or RabbitMQ) and Spring Cloud Stream, improving decoupling, scalability, and fault tolerance between microservices and the Vue frontend.
- Practiced Test-Driven Development (TDD) with JUnit and Vue Test Utils, and established a robust observability stack with Prometheus, Grafana, Elastic APM, and AWS CloudWatch, ensuring system health and fast issue resolution.

Tech Stack: Vue.js, Pinia (Vuex), TypeScript, SCSS, Java, Spring Boot, Spring WebSocket, Spring Cloud Stream, REST API, Spring Data JPA, PostgreSQL, WebSockets, Docker, AWS ECS, AWS Lambda, AWS S3, AWS CloudFormation, AWS CloudWatch, Apache Kafka, RabbitMQ, Prometheus, Grafana, Elastic APM, Postman, JUnit, Vue Test Utils, Git, CI/CD, TDD, Figma, SEO, WCAG

Sophos Solutions

5 years 6 months

Senior Web Developer

September 2015 – February 2018 (2 years 6 months)

- Spearheaded the development of a high-performance Single Page Application (SPA) for a financial services platform using Vue.js, Composition API, and Pinia (or Vuex), resulting in a 30% increase in user engagement and faster client-side interactions.
- Refactored a legacy monolithic architecture by decoupling the frontend and integrating it with scalable Java Spring Boot RESTful APIs, reducing server response times by 40% and improving maintainability through backend services.
- Enhanced frontend performance using advanced techniques including code splitting, lazy loading, tree shaking, and asset compression, reducing load times by 35% and boosting SEO rankings through structured metadata and schema markup (Open Graph, JSON-LD).
- Delivered fully responsive, mobile-first interfaces compliant with WCAG accessibility guidelines and optimized for Google AMP, leading to a 25% increase in mobile user traffic.
- Established a robust testing ecosystem using Vue Test Utils, Jest, Playwright, and Cypress, integrated within GitHub Actions–based CI/CD pipelines, which reduced post-deployment issues by 75% and improved release reliability.
- Partnered with designers using Figma and Adobe XD to revamp complex enterprise dashboards, improving navigation speed by 50% and increasing cross-functional team efficiency through a clean, modular design system.

Tech Stack: Vue.js, Composition API, Pinia (Vuex), TypeScript, HTML5, SCSS, Java, Spring Boot, Spring Data JPA, REST API, PostgreSQL, Redis, Vite, Webpack, GitHub Actions, Jenkins, Azure App Service, Playwright, Cypress, Jest, Vue Test Utils, Postman, WCAG, AMP, SEO, Google Analytics, Figma, Adobe XD, CI/CD, TDD

Web Developer

September 2012 – February 2015 (2 years 6 months)

- Developed responsive, accessible, and standards-compliant web applications using Vue.js, TypeScript, HTML5, and SCSS, adhering to modern usability principles and WCAG guidelines to broaden accessibility and improve user engagement.
- Engineered modular and reusable Vue components leveraging the Composition API, Vue Router, and Pinia (or Vuex) to enable a scalable frontend architecture aligned with evolving product requirements.
- Designed and integrated RESTful APIs using Java Spring Boot and Spring Data JPA, delivering secure, extensible backend services and structured content models to support complex application workflows.
- Streamlined frontend and backend build pipelines using Vite, Webpack, and Maven/Gradle, implementing environment-specific configurations, asset optimization, and release automation across development, staging, and production environments.
- Collaborated with backend engineers to implement Spring Boot–based caching strategies (e.g., Caffeine, Redis, and HTTP response caching), and deployed CI/CD pipelines via GitHub Actions and Jenkins, enabling faster, more reliable deployments.
- Optimized database performance in PostgreSQL by refactoring JPA queries, applying appropriate indexing strategies, and tuning ORM configurations, resulting in faster response times and improved backend throughput in high-traffic applications.
- Reduced frontend load times by 35% using advanced performance techniques such as code splitting, lazy loading, image optimization, and minification, boosting SEO visibility and ensuring a smoother user experience.

Tech Stack: Vue.js, TypeScript, HTML5, SCSS, Composition API, Vue Router, Pinia (Vuex), Java, Spring Boot, Spring Data JPA, REST API, PostgreSQL, Redis, Caffeine, Vite, Webpack, Maven, Gradle, GitHub Actions, Jenkins, Azure App Service, Google Analytics, SEO, WCAG, CI/CD, Postman, Jest, Playwright, TDD, Figma

EDUCATION

Universidad Católica de Pereira 04.2005 – 09.2012

Bachelor of Computer Science

SKILLS

Languages:	C++, JS(ES6), TS, Python, C#, Rust, PHP, Java, HTML5
Frameworks and Libraries:	Vue, Vuex, Nuxt, React.js, Next.js, Gatesby, Angular, Node.js, Express, React Native, Redux, RxJS, REST API, Django, JavaSpringBoot, Antd, GraphQL, Ably, Laravel, Tailwind CSS, RabbitMQ, Kafka, TypeORM, Shopify, WordPress
DBMS	MySQL, MS SQL, PostgreSQL, MongoDB, AWS DynamoDB, AWS RDS, Azure SQL Database
IDEs and Tools:	Visual Studio Code, Jenkins, Cypress, Jest, Jasmine, Terraform, Docker, Kubernetes, Redux DevTools, Babel, Webpack, Storybook, Azure DevOps, Git Bash, PowerShell
Cloud Services:	AWS Cloud Services, Microsoft Azure, GCP
Other skills:	Problem Solving, Full Stack Development, Critical Thinking, Leadership, Algorithm Design, Team-Oriented, SOLID, Agile (Scrum), Microservices, Test-Driven Development (TDD), Diagnosing, Business Analysts